SAMSUNG

AI Course

Capstone Project Final Report

For students (instructor review required)

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Together for Tomorrow! Enabling People

From Traditional Machine Learning to Transformers: Comparative Evaluation in Fake Review Detection and Topic Analysis

(07/01/2026)

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1. Introduction

1.1. Background Information

The project focuses on developing an artificial intelligence model aimed at detecting *fake reviews* and categorising the main topics addressed in user comments.

To achieve this objective, a dataset of reviews from Amazon and Yelp has been used to process comments related to products and companies, respectively. Based on the textual content of the comments, together with additional information associated with each review, machine learning and natural language processing models are trained and evaluated to identify linguistic and contextual patterns characteristic of misleading reviews, as well as to classify comments according to their theme or product category.

This makes it possible to gain a better understanding of user behaviour and the dynamics of content generation on e-commerce platforms. The project offers a useful tool for improving the reliability of published opinions, while supporting decision-making by consumers and businesses.

1.2. Motivation and Objective

In recent years, the exponential growth of e-commerce has made user reviews a key source of information for consumers and businesses. However, the use of fraudulent practices in the digital environment, such as writing fake reviews, has grown in parallel, potentially negatively influencing decision-making and reliability.

With the aim of enabling any user or entity to distinguish between them, various machine learning techniques are presented. In addition, as a complementary measure, the thematic classification of comments allows structured information to be extracted from large volumes of text, facilitating the analysis of trends, user preferences and relevant product characteristics.

1.3. Members and Role Assignments

Marina Benítez and Miguel Ruiz, students on the EOI's Artificial Intelligence Development course, are applying the knowledge they have acquired on machine learning and neural networks to this project, working in a collaborative environment with no defined roles.

1.4. Schedule and Milestones

The first tutorial took place on 15 November, during which the working method was defined and practical examples of different real-life implementations of artificial intelligence were presented as possible starting points. In the second tutorial, the main theme of the project was finalised, as well as the data collection methods to be used. Subsequently, in the third session, the different models developed and their respective results were presented and compared. Finally, in the fourth session the web page was successfully developed as part of the project.

In general, the tutorials were used both to share the progress of the project and organise each of its phases, and to support the understanding of the course syllabus. The more practical part of the project, from data acquisition to the implementation and development of the different models, was carried out independently, due to the long period of time required.

2. Project Execution

2.1. Data Acquisition

Once the main research topic had been defined, a search for information was conducted with the aim of obtaining a reliable and sufficiently broad dataset to make predictions.
On the one hand, the *Amazon Fake Reviews* dataset was obtained from Kaggle, focusing on products sold on the platform itself.
On the other hand, the Yelp Dataset was downloaded from the *ScienceDB* (Science Data Bank) repository in order to process comments in a different context, such as businesses.

2.2. Training Methodology

2.2.1 Data Preparation

The first step in the process was data preparation and cleaning. This stage is essential to ensure the quality of the dataset, as clean and well-structured data is essential for building an effective predictive model.

2.2.2 Model selection

Depending on each dataset and each expected objective, different models were used. From Logistic Regression, Support Vector Machines, Random Forest, among other supervised learning models, to deep learning and transformer models. This ensures the development of an accurate and reliable predictive model.

2.2.3 Training and validation

During the training phase, we will experiment with various machine learning models for detecting fraudulent reviews. The objective will be to determine the model with the best performance based on the characteristics of the dataset.

2.3. Workflow

The project was developed using collaborative workspaces such as GoogleColab, complemented by local teams for planning, testing, implementation, and model improvement tasks.

2.4. System Design

As mentioned above, the data was first acquired in such a way that it was suitable for the project. Although these are complete and reliable datasets, they need to be cleaned and prepared for use in the models. This includes removing null values, encoding categorical variables, and normalising features to ensure that the models receive high-quality data.

Once the data is ready, different models are trained to find the one that offers the best performance. The processed data is fed into the models so that they can learn the relationships and patterns that enable them to detect fake reviews.

The trained models are validated using metrics such as accuracy, recall, confusion matrix, etc. This phase is essential to determine which model performs best and provides the most accurate predictions.

With the model trained and validated, predictions are made to categorise new reviews. This step consists of applying the model to new data to generate the desired results.

3. Results

3.1. Data Preprocessing

To carry out this project, we worked with two different datasets: the *Amazon Fake Reviews* Dataset, which deals with reviews of different products, and the *Labelled Yelp* Dataset, which contains records of comments from different companies. In this way, the problem is approached from two completely different datasets, both in terms of size and context. Therefore, data pre-processing will depend on each dataset and will also be concatenated.

3.1.1. Pre-processing: Amazon Fake Reviews Dataset

The selected dataset consists of 40,432 reviews (rows), written in English and labelled: CG (*Computer generated*), OR (*Original reviews*). In addition, the *rating* column indicates the rating, out of 5, that the review has received from the user, and the *category* column allows us to identify the subject of the review.

As there is no null data, it is not necessary to process missing data. Similarly, there is no duplicate data or redundant columns. However, it is necessary to map the categorical values to numerical values: "OR:0", "CG:1", and Label Encoding is used to convert each unique category into an integer.

Another positive aspect in terms of detecting fake reviews is that the dataset is balanced, i.e., it has the same number of real reviews as "fake" ones.

3.1.2. Pre-processing: Yelp Dataset

In this case, the dataset is much larger, around 600,000 reviews, and has more columns that are unnecessary for the purpose of this project. Thus, the repeated and irrelevant columns are first removed, and then the categorical values are remapped to numerical values, maintaining the relationship: 0=real, 1=fake.

However, it was detected that training with the original dataset caused the model to systematically predict the majority class, ignoring the minority class and distorting the performance metrics, as it was heavily biased towards the majority class (89% - 11%).

To solve this problem, a random *undersampling* technique was applied. All samples from the minority class were retained, and an identical number of samples from the majority class were randomly selected. This resulted in a balanced dataset with a 50/50 distribution and a total size of approximately 160,000 samples, eliminating the inductive bias towards the "Fake" class.

3.1.3. Concatenation

In order to observe how the different models developed with both datasets behave at the same time, they are concatenated. There must be consistency between the two, specifically, the number of columns must be the same and they must have the same name.

3.1.4. Pre-processing for Topic Modelling (BERTopic)

The *YelpZip* dataset was used to analyse topics in restaurant complaints. Processing focused on reducing semantic noise:

> **1. Filtered by Sentiment**: Since the objective was to analyse complaints, the original dataset of 608,458 reviews was filtered, retaining only those with a rating of 2 stars or less. We then worked with a subset of 78,577 documents with high negative relevance.

2. Removal of Specific Stop Words: In addition to sklearn's standard English stop words, a custom blacklist was created to remove generic hospitality terms that do not contribute distinctive semantic value (e.g., "food," "place," "service," "restaurant," "order," "time," "staff"). This allowed the BERTopic model to focus on the nouns and verbs that truly differentiate a problem (e.g., "cold pizza") from a generic review.

3. Vectorisation: CountVectorizer was used with an n-gram range of (1, 2) to capture both individual words and word pairs.

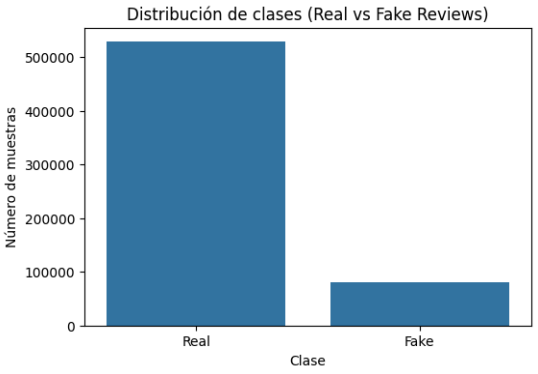
3.2. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a crucial stage in which the dataset is examined to identify patterns, trends, and potential problems, enabling informed decisions to be made during the pre-processing and model selection phases. Using descriptive statistics and visualisations, the differences between real and fake reviews are explored, and relevant characteristics of the dataset are identified.

Exploratory analysis was essential in identifying the unfeasibility of training with raw data in the classification task.

3.2.1. Diagnosis of Imbalance in Fake Reviews

Analysis of the target variable *(Label*) in both datasets revealed an extreme imbalance in the case of the original *Yelp* dataset:



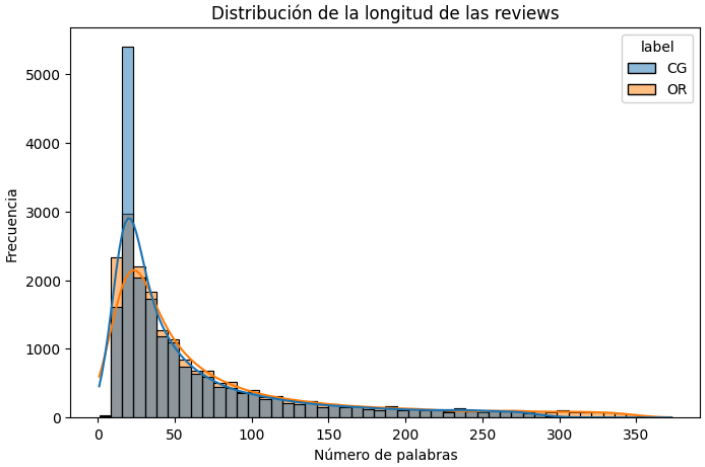
Impact on the Initial Model:

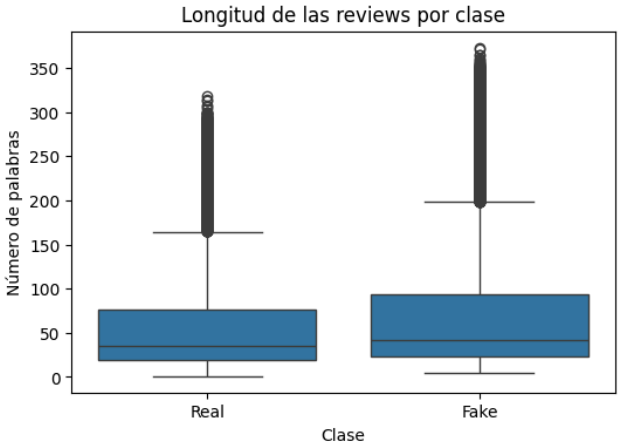
As an example of the impact of imbalance, the RoBERTa model quickly converges to an *accuracy* of 0.8955, matching almost exactly the proportion of the majority class. This indicated that the model was not learning linguistic patterns, but rather opting for a "lazy" strategy of classifying everything as "Real" to minimise overall error. This finding justified the immediate implementation of *the undersampling* described in section 3.1.2 to force the model to learn distinctive features.

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3.2.2. Review Length Analysis

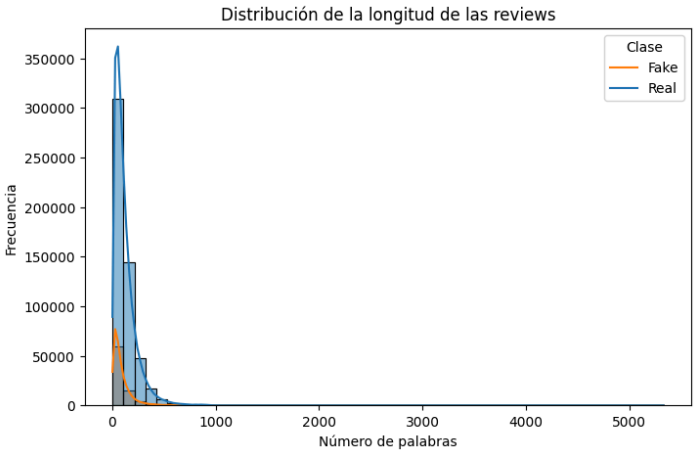
Amazon Dataset:

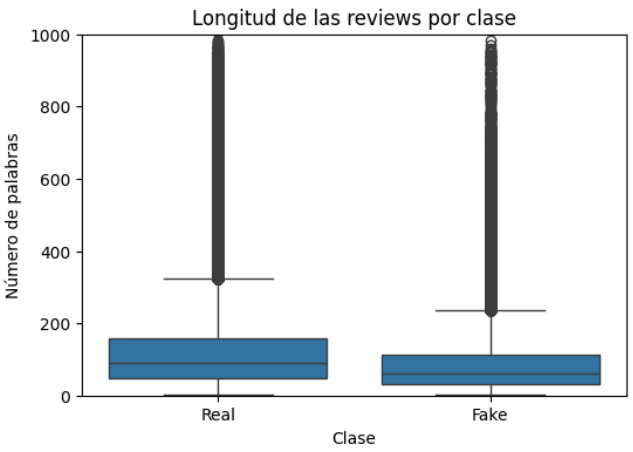




In general, fake reviews have a slightly higher median and greater dispersion in the number of words, as well as a greater number of outliers. This suggests that fake reviews tend to be longer and more variable, possibly due to attempts at persuasion or automatic text generation.

Yelp Dataset:

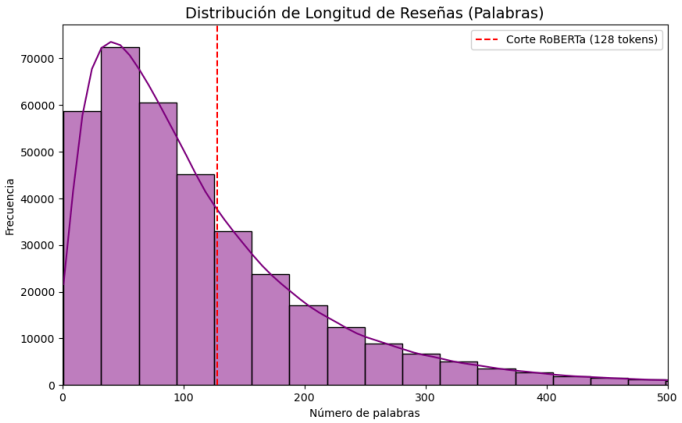




In this case, it is the real reviews that have a slightly higher median and greater dispersion in the number of words, being more extensive.

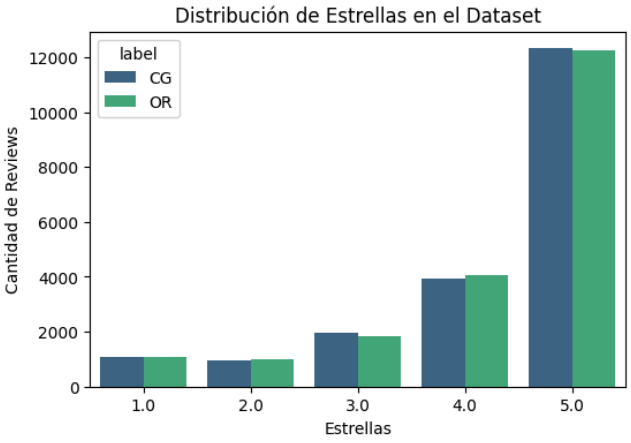
These differences lead to the conclusion that it is justified to consider text length as a relevant feature during the modelling phase. Although it is not a feature that is obvious at first glance, it can be useful at the algorithm level.

To correctly configure the **RoBERTa** model, it was necessary to analyse the length of the reviews, as Transformer models have an input limit (typically 512 tokens); in fact, processing very long sequences is computationally expensive. Analysis of the word distribution showed that the vast majority of reviews are concentrated below 150 words. This justified the choice of MAX_LEN = 128 for the tokeniser, covering most of the semantic content without wasting GPU memory on unnecessary *padding.*

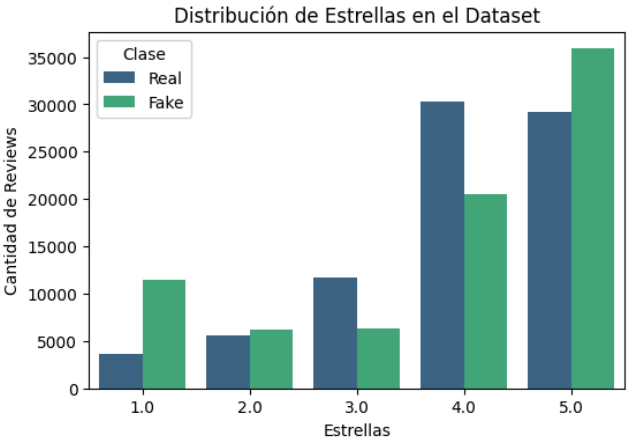


3.2.3. Rating Vs Label

Amazon Dataset:



Yelp Dataset:



It can be seen that no clear conclusion can be drawn regarding the relationship between rating and review accuracy in general. However, it can be noted that in the Yelp dataset there are a greater number of fake reviews at the extremes, 1 and 5 stars, which makes sense given their purpose.

3.2.4. Semantic Structure of Complaints (Topic Modelling)

Analysis of the 78,577 negative reviews using dimensionality reduction (UMAP) and clustering (KMeans on *all-MiniLM-L6-v2* embeddings) revealed a clear structure in user complaints, divided into two macro-categories:

1. **Management and Service Issues (Clusters 0 and 2):**
    - ○ These are the most voluminous topics.
    - ○ Keywords: *minutes, table, wait, asked, told, rude, manager*.
    - ○ These represent complaints that are common to any type of restaurant, focusing on waiting times and staff treatment.

2. **Product-Specific Issues (Clusters 3 to 14):**
    - ○ The remaining topics group reviews by type of cuisine, indicating that complaints about food tend to be very specific to the dish.
    - ○ Examples detected: Topic 7 (Pizza): *crust, slice, pie, cheese*. Topic 9 (Mexican): *tacos, burrito, salsa, guacamole*. Topic 13 (Japanese): *sushi, rolls, rice, fish*.

3.3. Modelling

To obtain the best results, a wide variety of models have been implemented, playing with the corresponding parameters. In this way, both datasets have been studied separately, Amazon and Yelp, as well as their concatenation.
Thanks to GridSearchCV, each model has been adjusted, ensuring cross-validation to avoid sample bias and find the best combination of hyperparameters.

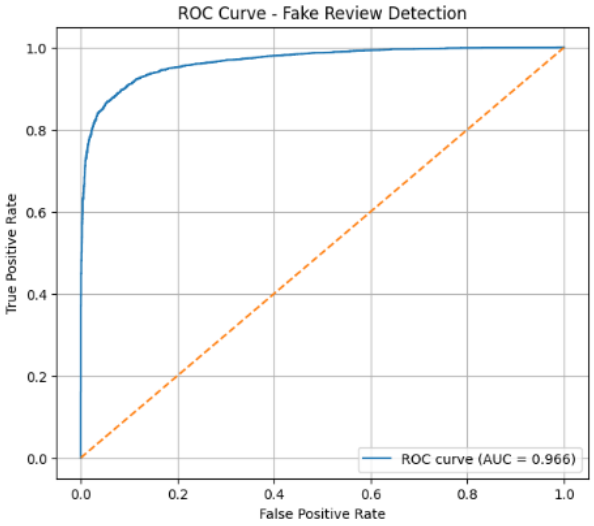
3.3.1. Amazon Dataset

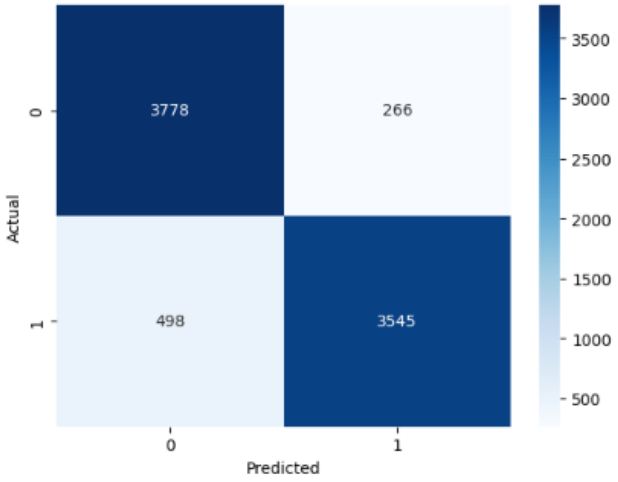
This dataset consists of a significantly smaller number of reviews than the Yelp dataset, so it was used primarily for testing, as its execution speed and computational cost are lower. The data is labelled, i.e. supervised learning models are used to correctly predict the output of new data.

1. Logistic Regression

We begin by testing a simple and interpretable model such as Logistic Regression. This model allows us to estimate the probability that a review is fake and is a good starting point before using more complex models.

In this case, TF-IDF (Term Frequency – Inverse Document Frequency) has been used to convert the text into numbers, as it is usually more effective than BoW because it helps to highlight words that are distinctive in fake and real reviews. In addition, Pipeline is used to prevent data leakage and improve organisation.



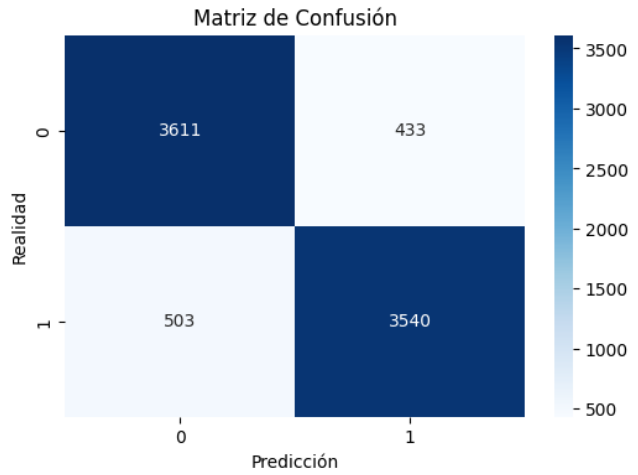


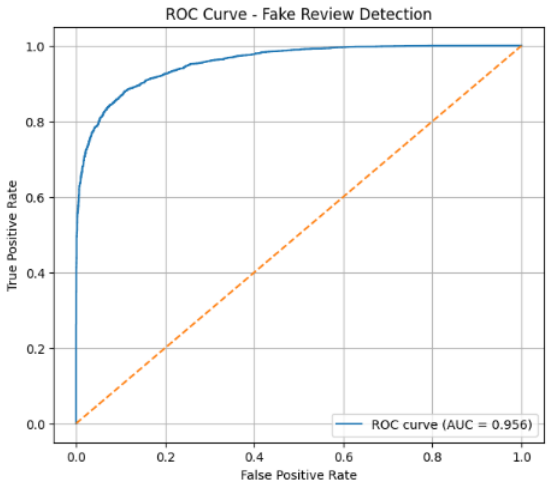
Additionally, we have tried changing the model type from PLN to Word2Vec, a word embedding method that transforms each word into a dense numerical vector, capturing semantic and contextual relationships. However, in this case, TF-IDF works better because it better captures repetitive words, which are key in fake reviews.

In conclusion, this study demonstrates that relatively simple approaches, such as TF-IDF combined with Logistic Regression, can offer robust and reliable results in detecting fake reviews, achieving a performance of 0.91. This result confirms that the importance and relative frequency of words are key factors for this type of problem, and that well-regularised linear models can offer outstanding performance without the need for more complex architectures.

2. Multinomial NB

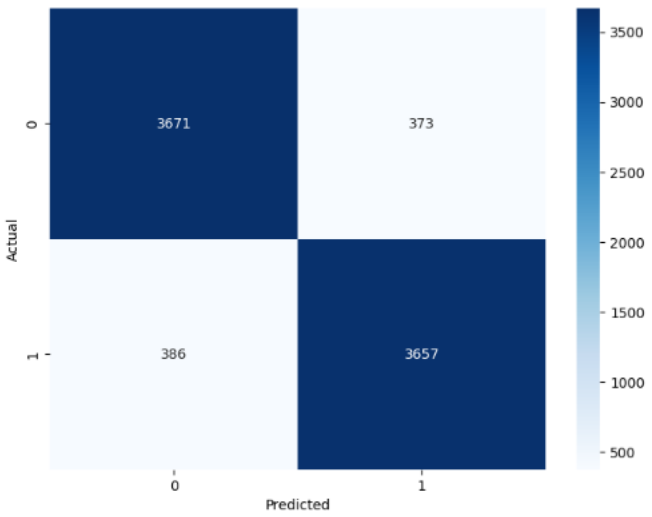
The Multinomial Naive Bayes model was evaluated as a baseline model due to its efficiency and suitability for textual data represented by TF-IDF. Although its performance was slightly lower than that of Logistic Regression, it offered competitive results with significantly lower computational cost, confirming its usefulness as a solid reference model for detecting fake reviews.





3. Support Vector Machines

Support Vector Machines (SVM) were evaluated due to their high performance in high-dimensional spaces and text classification tasks. When combined with TF-IDF features, LinearSVC obtained competitive results, making it a solid alternative to Logistic Regression. In fact, the confusion matrix confirms a low number of false negatives and false positives, demonstrating that the model is robust and reliable for detecting fraudulent reviews.
The hyperparameter C=10 indicates that a less regularised margin improves the model's ability to separate classes in the TF-IDF feature space.



On the other hand, SVM has been combined with embeddings, specifically Word2Vec, which works better since words with similar meanings (e.g. *good*, *excellent*) are close together in the vector space, unlike TF-IDF, which only sees words as indices, with no relationship between them. It should be noted that the execution time increases considerably.

	precision	recall	f1-score	support
0	0.93	0.90	0.91	4044.00
1	0.90	0.93	0.92	4043.00
accuracy	0.92	0.92	0.92	0.92
macro avg	0.92	0.92	0.92	8087.00
weighted avg	0.92	0.92	0.92	8087.00

4. Deep Learning

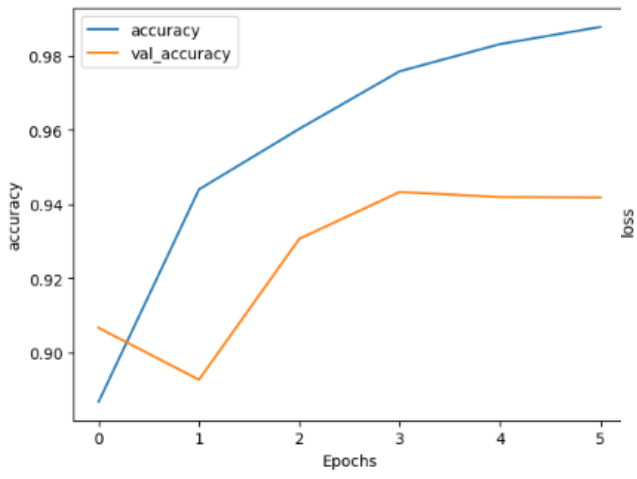
In order to achieve a greater leap in accuracy of the model, it is necessary to switch to deep learning techniques. They capture complex patterns and dependencies between words, and in general they excel at handling context and semantics better than TF-IDF or average Word2Vec.

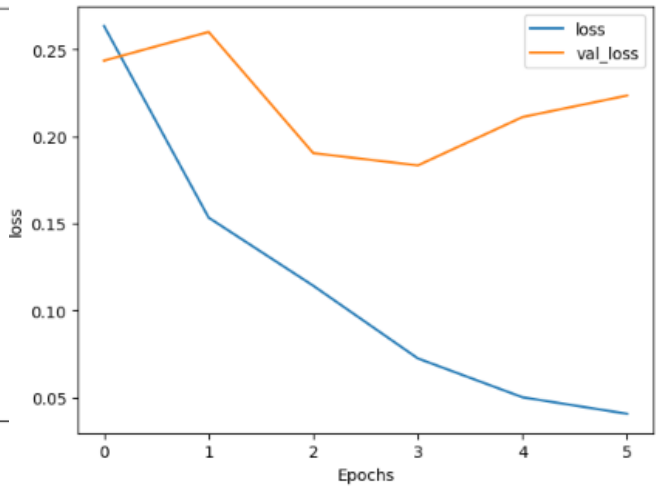
First, tokenisation is performed, converting words into indices, and padding, so that all sequences are the same length. Once the data is prepared, it is grouped into three subsets: training, testing and validation, and the model is defined, i.e. the different neural layers.

After conducting multiple tests, combining Embedding and Bidirectional LSTM has yielded an accuracy of close to 95%, making the extra computational expense worthwhile. However, the model is sensitive to overfitting.

	precision	recall	f1-score	support
0	0.94	0.95	0.95	4044
1	0.95	0.94	0.95	4043
accuracy			0.95	8087
macro avg	0.95	0.95	0.95	8087
weighted avg	0.95	0.95	0.95	8087

```
--- Matriz de Confusión ---
[[3852  192]
 [ 244 3799]]
```





5. SVM + LSTM

The best result for the Amazon dataset was obtained by averaging the predictions of the two best models: SVM and LSTM. SVM with Word2Vec looks at the "average" semantic meaning, while LSTM looks at the sequence and order of words. In this way, this mathematical "debate" between the two is what makes the combined system more robust than the separate models.

```
              precision    recall  f1-score   support

           0       0.95      0.95      0.95      4044
           1       0.95      0.95      0.95      4043

    accuracy                           0.95      8087
   macro avg       0.95      0.95      0.95      8087
weighted avg       0.95      0.95      0.95      8087

[[3850  194]
 [ 212 3831]]
```

Result of the ensemble with some example comments:

Text: 'I absolutely love this product, the quality is amazing and shipping was fast.'

- SVM confidence: 0.2734
- LSTM confidence: 0.7022
- Average: 0.4878
-> PREDICTION: CLASS 0 (Actual)

Text: 'Do not buy this rubbish. The box was empty and customer service is a scam.'

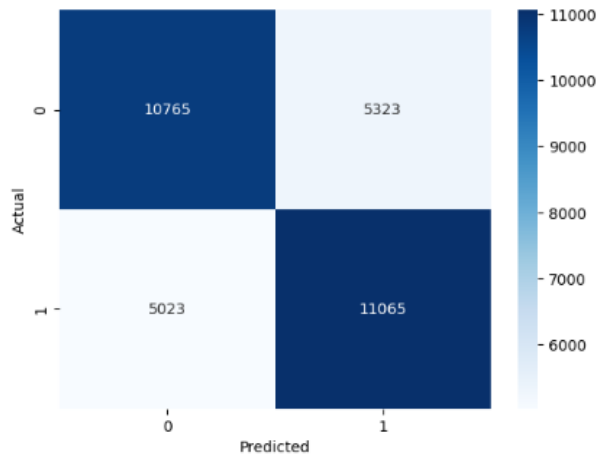
- SVM confidence: 0.0613
- LSTM confidence: 0.1458
- Average: 0.1035
-> PREDICTION: CLASS 0 (Actual)

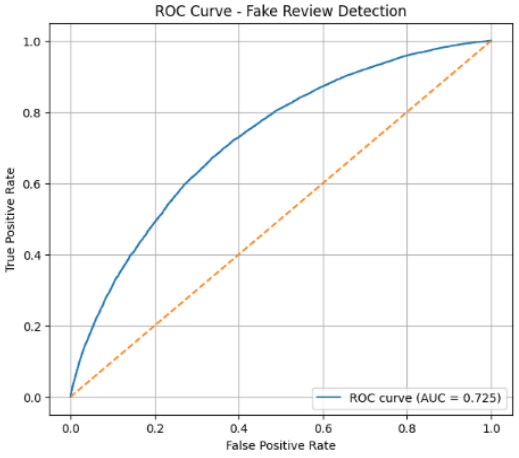
The problem with this dataset is that, despite its high accuracy, the model is restricted to the characteristics of the comments in it. In other words, if the reviews are old, it will be able to detect fake reviews generated by computers from that era, but it will not be very reliable today.

3.3.2. Yelp Dataset

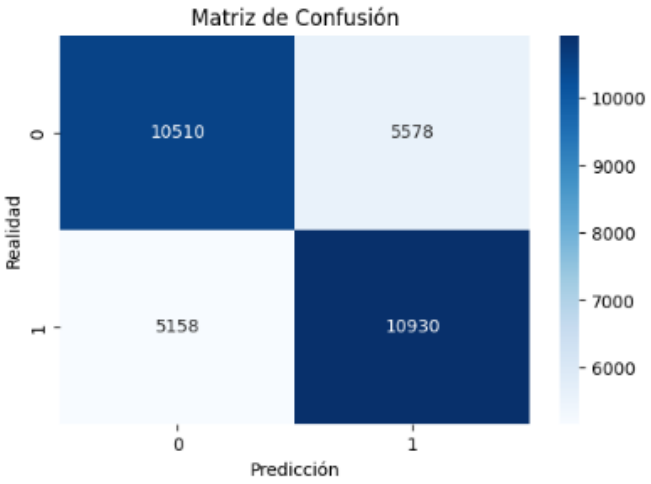
This dataset is more complex when it comes to detecting fake reviews because they are generated by humans, so performance will be negatively affected by the complexity of detection. However, in these cases, the use of classification algorithms is particularly useful, as anomalies are difficult to detect by human eyes.

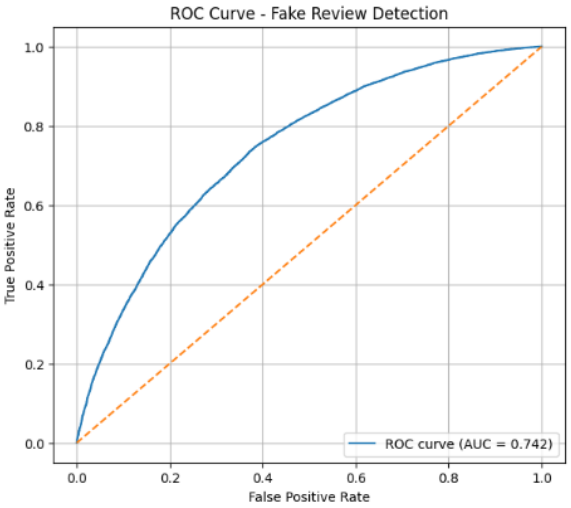
1. Logistic Regression





2. Multinomial NB





In both cases, TF-IDF (Term Frequency – Inverse Document Frequency) was used to convert the text into numbers, together with the combination of optimal hyperparameters in each case. The drastic decrease in accuracy mentioned above can be observed. For this reason, this dataset must be approached using new tools such as data engineering, taking into account known variables such as rating or user_id.

3. Ensemble Learning

Random Forest and XGBoost are ensemble learning algorithms widely used for classifying fake reviews, as they combine multiple decision trees and achieve high accuracy in fraud detection and text analysis problems. Random Forest uses bagging, while XGBoost uses boosting, so it is a good strategy to compare them.

After testing different NLP techniques (TF-IDF/Word2Vec), the best results were obtained with Random Forest, together with feature engineering: studying the deviation of the individual comment rating from the average for the product and user activity, data provided by the dataset.

```
Precisión: 0.732
              precision    recall  f1-score   support

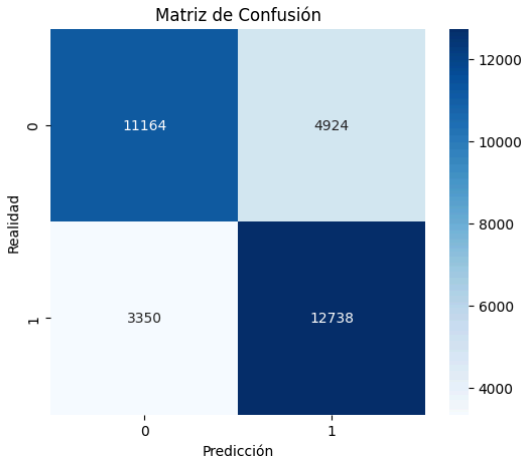
          -1       0.72      0.77      0.74     16174
           1       0.75      0.70      0.72     16002

    accuracy                           0.73     32176
   macro avg       0.73      0.73      0.73     32176
weighted avg       0.73      0.73      0.73     32176


Importancia de las nuevas variables:
rating: 0.0099
deviation: 0.0175
user_activity: 0.0808
```

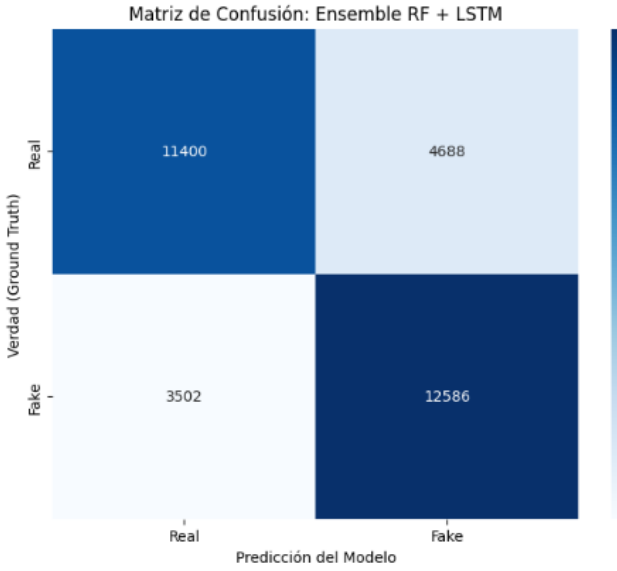
4. Deep Learning

As with the Amazon dataset, the best results are obtained with LSTM. The analysis focuses on textual content, as it models sequential dependencies in language, i.e., capturing context.



	precision	recall	f1-score	support
0	0.81	0.65	0.72	16088
1	0.71	0.85	0.77	16088
accuracy			0.75	32176
macro avg	0.76	0.75	0.74	32176
weighted avg	0.76	0.75	0.74	32176

5. Random Forest + LSTM
The best result for the Yelp dataset was achieved by averaging the predictions of the two best models: Random Forest and LSTM




```
Accuracy Ensemble: 0.7455
ROC AUC Score:     0.8214
------------------------------------------------
Reporte de Clasificación del Ensemble:
              precision    recall  f1-score   support

           0       0.76      0.71      0.74     16088
           1       0.73      0.78      0.75     16088

    accuracy                           0.75     32176
   macro avg       0.75      0.75      0.75     32176
weighted avg       0.75      0.75      0.75     32176


--- RESUMEN DE MEJORA ---
Random Forest solo: 0.7325
LSTM solo:          0.7419
Ensemble (Unión):   0.7455
```

6. RoBERTa

RoBERTa (*Robustly Optimised BERT Approach*) was selected for the detection of fake reviews. This architecture optimises BERT training by eliminating the prediction of the next sentence and using dynamic masking.
Justification: Detecting *fake reviews* requires identifying stylistic and semantic subtleties, not just grammatical errors. RoBERTa offers a more robust representation of context than standard BERT, which is critical for correct generalisation after *fine-tuning.*

3.3.3. Concatenation

In order to evaluate the model's generalisation ability and increase the volume of training data, the balanced Yelp dataset was concatenated with the Amazon Reviews dataset. The result obtained maintained an accuracy of approximately 0.75, similar to the performance of the model trained exclusively with Yelp.

By exposing the model to a more extensive and varied vocabulary, the neural network is forced to learn structural patterns of deception rather than memorising specific keywords for a topic. Furthermore, increasing the sample size reduces the likelihood that the model will memorise the peculiarities or statistical noise of the Yelp dataset. All in all, a model trained across multiple domains is theoretically better suited for deployment in unknown real-world environments, as it does not rely exclusively on the context of restaurants or local services.

However, the different semantic distribution between datasets is the main reason why accuracy did not increase significantly, as "semantic noise" is introduced that can confuse the classifier, diluting the specific signals that detect fraud on Yelp.
It should be noted that both the computational cost and execution time increase considerably.

It can therefore be concluded that, although the merging of datasets increased the training volume, the phenomenon of domain shift and the incompatibility of user behaviour metrics between platforms prevented a net improvement in accuracy. This confirms that the characteristics of review fraud are highly context-dependent.

3.3.4 Topic Classification

1. Supervised Learning

In order to analyse **multi-class classification**, rather than the binary classification addressed above, and compare the performance of different models in this task, the ***Amazon Fake Reviews*** dataset was used. This dataset is particularly suitable for the study of supervised classification, as each review is pre-labelled with the category to which it belongs.

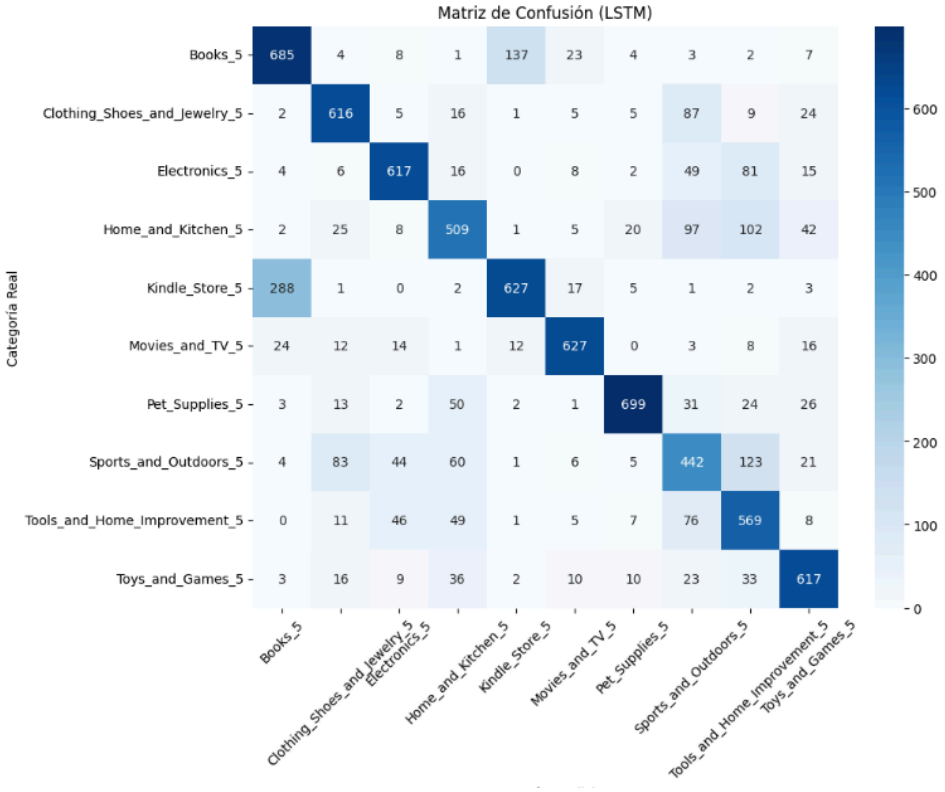


First, a model based on **Long Short-Term Memory (LSTM)** recurrent neural networks was used. After multiple experiments in which both the hyperparameters and the network architecture were varied, as well as the application of different techniques to mitigate *overfitting*, an overall accuracy of 73% was obtained.

However, when analysing the confusion matrix, it was observed that the model had difficulty differentiating between two specific categories: **Books** and **Kindle Store**. This behaviour is understandable, given that both categories share very similar vocabulary, which makes it difficult to discriminate based solely on linguistic patterns learned by the model.

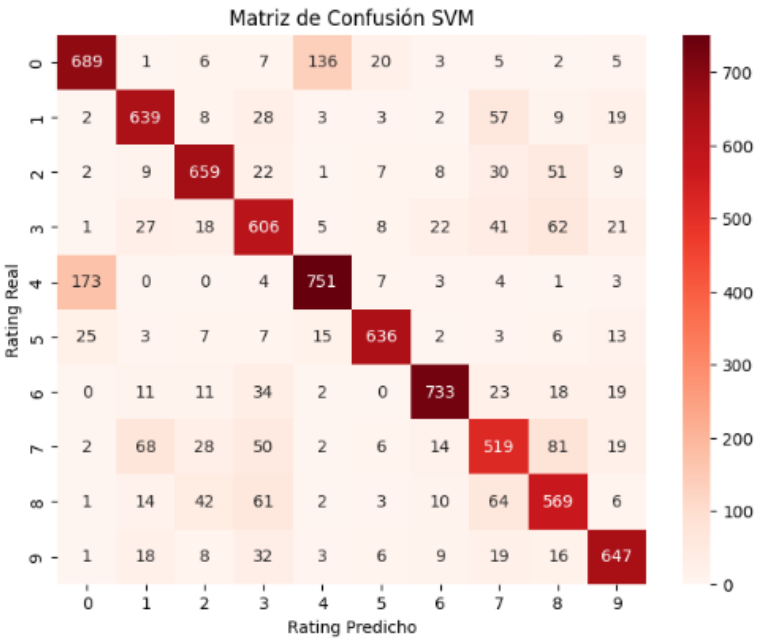
In order to improve performance in this distinction, two lists of keywords representative of each of these categories were introduced. These lists were used as additional information by means of an occurrence counter, providing the model with a complementary signal to facilitate the identification of each topic.

As expected, incorporating this additional information improved the model's performance, especially in classifying the aforementioned categories, demonstrating the usefulness of combining deep learning approaches with domain-specific knowledge.



Subsequently, a **Logistic Regression** model was used as an alternative to the LSTM-based approach. Despite its lower complexity, this model performed better, improving on the results previously achieved, requiring lower computational cost and faster training.

Finally, a model based on **Linear Support Vector Classifier (Linear SVC)** was evaluated, which achieved slightly better performance than both the LSTM model and Logistic Regression. This result confirms the suitability of linear classifiers for text classification tasks, especially when using **TF-IDF-based** representations and working in high-dimensional spaces. Linear SVC demonstrated a greater ability to separate the classes, even in cases where there is significant overlap in vocabulary between categories.



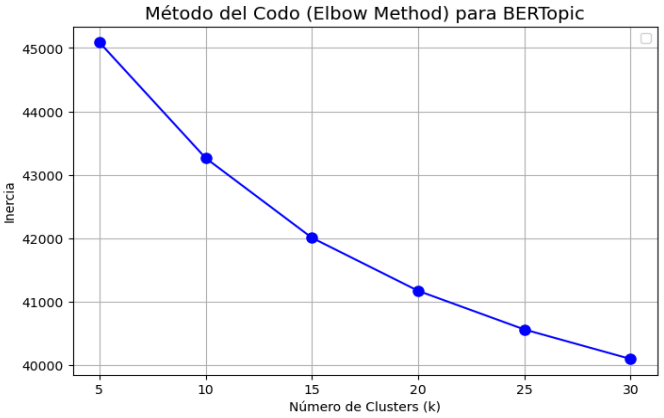

```
--- PRUEBA FINAL ---
Texto: The mobile phone battery is amazing
Predicción: Electronics_5
------------------------------
Texto: I love running with these shoes
Predicción: Clothing_Shoes_and_Jewelry_5
------------------------------
Texto: The frying pan is non-stick
Predicción: Home_and_Kitchen_5
```

2. Unsupervised Learning: BERTopic

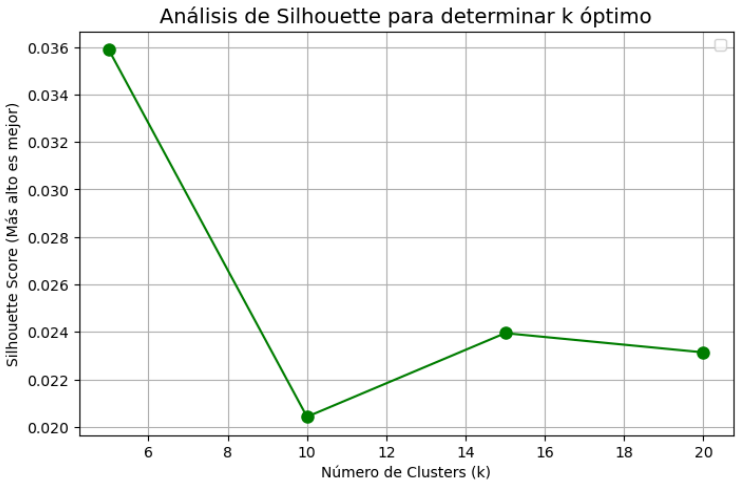
BERTopic, which combines transformer *embeddings* with clustering algorithms, was used to extract topics from negative reviews.

Clustering Adaptation (From HDBSCAN to K-Means):
The standard implementation of BERTopic uses HDBSCAN to cluster documents. However, in this dataset, this algorithm classified an excessive percentage of reviews as "noise" (outliers, class -1), losing valuable information about minority but relevant complaints.

Therefore, it was decided to replace it with **K-Means**. To determine the optimal number of themes, cluster inertia was analysed (Elbow Method). Although the curve does not show a perfect elbow due to the complexity of natural language, k=15 was selected as the equilibrium point where granular themes (distinguishing types of food) are obtained without excessively fragmenting complaints about the service.



As can be seen in this graph, there is no elbow-shaped curve, so we relied on the Silhouette analysis, which perfectly complements the Elbow Method.



Here we see that 5 clusters would be the optimal number according to the Silhouette analysis, but in that case we lose many topics for each type of restaurant (pizza, hamburgers, sushi, etc.) since it groups together topics that talk about food, and in this case we think it is more interesting to separate them by type of review according to the food. So we chose the second best, which is 15.

3.4. User Interface

To bridge the gap between the theoretical development of the model and its practical application, a functional web interface prototype was developed. The aim of this module is to democratise access to the detection system, allowing users without technical knowledge to evaluate the veracity of a review in real time.

3.4.1. Web System Architecture

The application was built using the **Flask** micro-framework (Python). This choice was based on its lightness.
The architecture follows a simple client-server pattern:

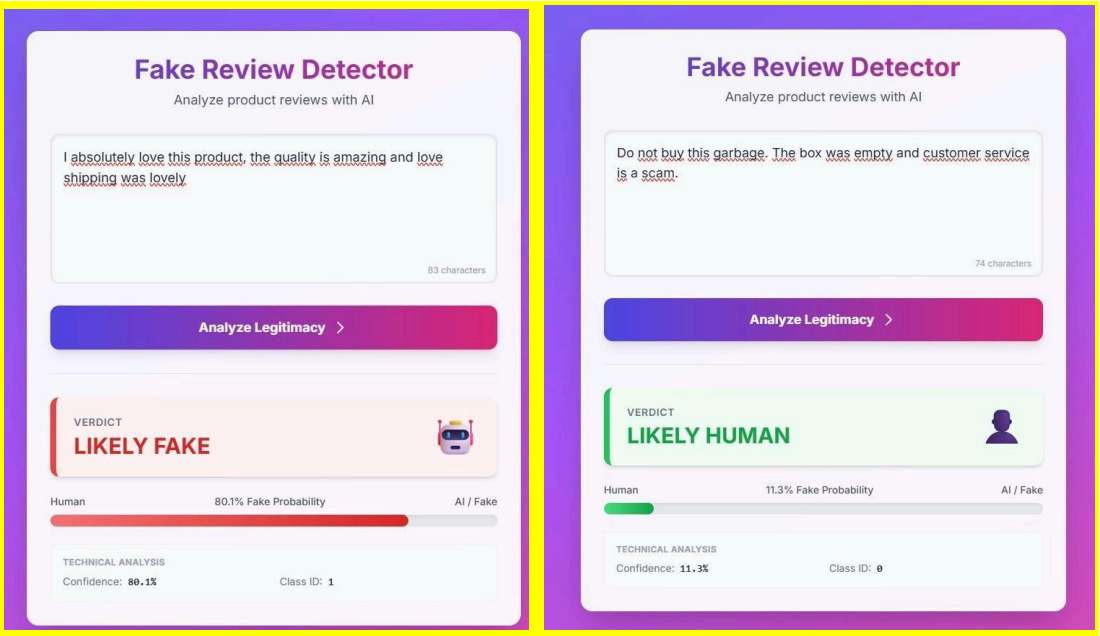
1. **Backend (API):** A REST *endpoint* (/predict) was exposed to receive the raw text, execute the pre-processing pipeline, and perform inference using the weights of the trained model.
2. **Frontend:** A minimalist interface was designed using HTML and CSS, with a focus on usability.

3.4.2. Interaction Flow and Results Visualisation

The interface allows the user to enter the text of a suspicious review into a text area and submit the analysis request.

Once the input has been processed, the system does not simply offer a binary classification (Real vs. Fake), but provides a higher level of granularity through a **confidence score confidence score**.

- **Transformation from Logits to Probability:** The raw output of the model (logits) is passed through a *Sofitmax* function, converting the numerical values into probability percentages that add up to 100%.
- **Final Result:** The application shows the user the predicted label (e.g. *"Computer-generated"*) along with the certainty percentage (e.g. *"98.5%"*). This is crucial for decision-making in borderline cases where the model may not be entirely certain (probabilities close to 50%).



3.5. Testing and Improvements

This section details the performance evaluation of the two implemented models.

3.5.1. Topic Modelling Evaluation (BERTopic)

As this is unsupervised learning, the evaluation of BERTopic focused on the consistency and interpretability of the clusters generated.

- Qualitative Results: Inspection of the resulting topics confirmed high quality in semantic separation. The model was able to clearly distinguish between operational complaints (Clusters 0 and 2: waiting times, service, rudeness) and product complaints (Clusters 3-14: pizza, sushi, Mexican food), validating the model's usefulness in understanding the causes of customer dissatisfaction without the need for manual labelling.

3.5.2. Classifier Evaluation (RoBERTa)

Training the *fake review* detection model required fine-tuning of hyperparameters to achieve convergence.

- **Convergence Adjustment:** Initial tests yielded an accuracy of 50% (random), indicative of a destructive learning rate. Correcting the *learning rate* to 2e-5 and unfreezing the base layers allowed the model to begin to learn real patterns, overcoming the initial blockage.
- **Final** *Performance***:** The model reached its peak performance in **Epoch 2**, with a validation accuracy of **68.87%** and the lowest error recorded (*Loss*: 0.5950). This demonstrates that, with balanced data, the model is capable of distinguishing fake reviews with significantly higher reliability than random chance.

Overfitting Diagnosis and Future Optimisations:

Analysis of the learning curves revealed that, starting in the third epoch, the model began to suffer **from overfitting**: while the accuracy in the training set continued to improve (exceeding 74%), the validation error increased (from 0.59 to 0.62). This indicates that the model began to memorise specific examples from the training set rather than generalising linguistic rules.

To mitigate this behaviour and improve the robustness of the model in future implementations, the following architectural improvements are proposed:

1. **Early Stopping:** Implement a control mechanism that automatically stops training and saves the model weights at the exact moment when validation loss stops decreasing, preventing the performance degradation observed in later epochs.
2. **Regularisation through Weight Decay:** Integrate an L2 penalty into the AdamW optimiser. This restricts the magnitude of synaptic weights, preventing the model from assigning disproportionate importance to specific words that could be statistical noise.
3. **Increased Dropout Rate:** The base configuration of RoBERTa uses a *dropout* of 10%. Increasing this probability (e.g., to 20% or 30%) in the attention and hidden layers would force the network to learn more distributed and redundant representations, reducing dependence on specific neurons and improving generalisation to unseen texts.

4. Projected impact

This section analyses the potential impact of the developed system, highlighting the technical and functional milestones achieved, as well as the roadmap for the project's evolution towards a robust production environment.

4.1. Accomplishments and Benefits

Overall, the project brought clear benefits at both the academic and practical levels. The main objectives of the project have been successfully achieved: the detection of fraudulent reviews and the categorisation of topics. In this way, the knowledge acquired in the course has been applied and further developed.
Despite the limitations of the datasets used and the models themselves, satisfactory results have been obtained, as well as a comparative study of the models employed.
In addition, data visualisation libraries were used to generate clear and revealing graphical analyses, facilitating better interpretation and communication of the results.

Complementing the course content, *transformers* were used as they are very powerful tools in natural language processing. The benefits and challenges that have been overcome with these models are outlined below:

1. **Class Bias Resolution:** The "accuracy paradox" in the detection of fake reviews was identified and neutralised. Through *undersampling* techniques and fine-tuning of hyperparameters in **RoBERTa**, we were able to transform a model that initially acted randomly (50%) or with bias (90% artificial) into a competent classifier capable of distinguishing subtle linguistic patterns with an accuracy of nearly **69%** in a balanced environment.
2. **Robust Semantic Segmentation:** Adapting the **BERTopic** model by replacing HDBSCAN with **K-Means** allowed us to recover critical information that was previously discarded as noise. This resulted in a clear taxonomy of 15 complaint categories, validating the system's ability to structure unstructured data without human supervision.
3. **Reputation and Trust** Protection**:** The system provides a barrier against smear campaigns (false negative reviews) or fraudulent self-promotion, increasing consumer confidence in review platforms.
4. **Automated Operational Intelligence:** For restaurant owners, the *Topic Modelling* module automates the reading of thousands of complaints. Instead of manually processing text, the system provides immediate diagnostics: it allows you to distinguish whether a drop in ratings is due to operational problems (e.g. 'Waiting times', 'Staff attitude') or product deficiencies (e.g. 'Cold pizza', 'Spoiled sushi'), facilitating specific corrective decision-making.

4.2. Future Improvements

Despite promising results, the transition from this prototype to a large-scale production system requires addressing certain limitations observed during the experimental phase.

As has been demonstrated, one of the most important factors is the dataset used. In practice, better results have been obtained with a very specific dataset on a given topic, but more important is the content of the reviews. The fact that the fake Yelp reviews were written by people has made the model less efficient. But, at the same time, it is in these cases that Artificial Intelligence is essential, as they are not detected by human perception.

In conclusion, in the future, we could delve even deeper into other more complex machine learning techniques, such as *transformers,* train models with much more recent data, study the impact of other variables on training and prediction, among others.

Supervised Model Optimisation (RoBERTa):

- **Overfitting Mitigation:** Given that the model tends to overfit from the third epoch onwards, it is a priority to implement advanced regularisation techniques, such as *Weight Decay* and an increase in the *Dropout* rate.
- **Data Augmentation:** To improve generalisation without reducing the dataset (as occurs with *undersampling*), we propose using data augmentation techniques such as back translation or the generation of synthetic paraphrases for the minority class (real reviews), allowing training with a larger volume of balanced data.

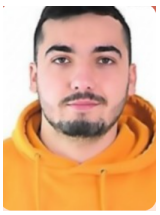
Evolution of the Unsupervised Model (BERTopic):

- **Dynamic Clustering:** Although K-Means (k=15) offered solid results, it requires defining the number of topics a priori. A future improvement would be to implement **hierarchical clustering** algorithms that allow topics and subtopics to be discovered dynamically without predefining their quantity, adapting to the natural evolution of complaints over time.

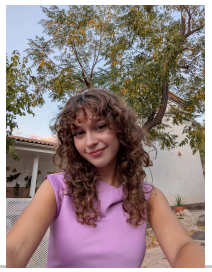
Functional and Deployment Improvements:

- **Explainable AI (XAI):** To increase adoption of the system, it is crucial to implement interpretability libraries such as **LIME** or **SHAP**. This would allow the end user to understand *why* a review has been classified as fake (e.g., by highlighting suspicious words or structures), avoiding the "black box" effect.
- **Optim**isation **for Inference (Quantisation):** To reduce latency in the web application and computational cost, it is recommended to apply **quantisation** techniques (reducing the precision of weights from 32 to 8 bits) or *knowledge* distillation to deploy lighter models such as *DistilRoBERTa* with similar performance.

5. Team Member Review and Comment



Miguel Ruiz Estudillo



Marina Benítez Sánchez

NAME	REVIEW and COMMENT
Miguel Ruiz Estudillo	This project has been an excellent opportunity to go beyond the limits of the academic syllabus. The implementation of **Transformer** models (RoBERTa and BERTopic) and the resolution of complex problems, such as class imbalance, have been a very motivating technical challenge. It has been a very good experience to understand how to apply cutting-edge technologies to real-world problems.
Marina Benítez Sánchez	Thanks to my participation in this project, I was able to apply and further develop the knowledge acquired during the Artificial Intelligence course, implementing the studied models and exploring each of them in depth in order to achieve the best possible results. Furthermore, this project allowed me to appreciate one of the many practical applications of Artificial Intelligence in real-world scenarios and to understand how this powerful tool can support us in solving everyday problems and addressing real-life challenges.

6. Instructor Review and Comment

CATEGORY	SCORE	REVIEW and COMMENT
IDEA	/10	
APPLICATION	/30	
RESULT	/30	
PROJECT MANAGEMENT	/10	

PRESENTATIO N & REPORT	/20	
TOTAL	/100	